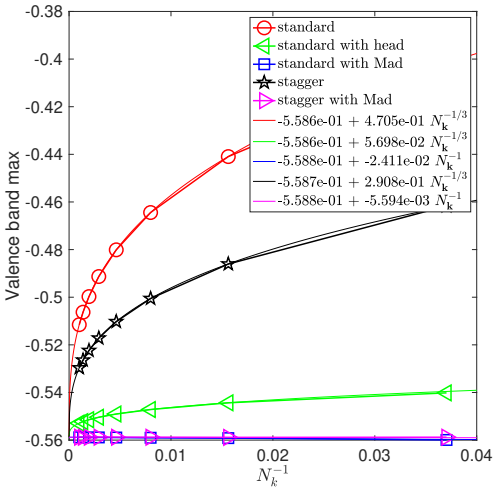
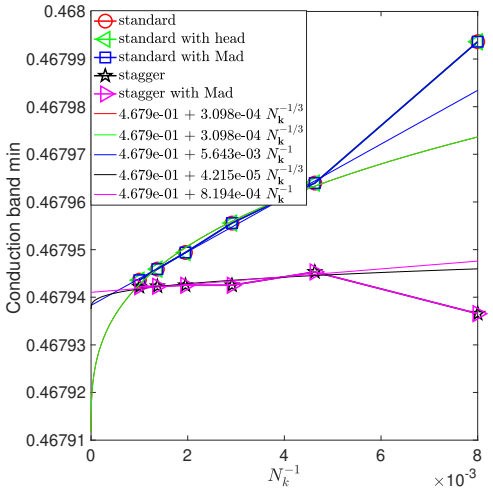
Experiment setting: H2, gth-szv, kecut=100, number of frequency points = 100, omega cutoff = 1.0, linearized G0W0, $\Sigma = \Sigma^c + \Sigma^x - V_{xc}$

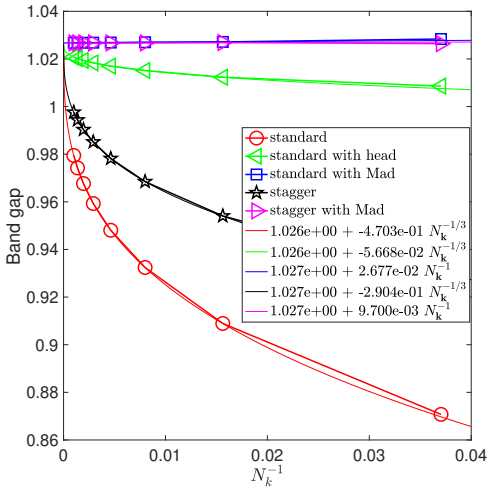
Setting 1 only include $\Sigma^x - V_{xc}$



(a) VBM at $\mathbf{k} = \mathbf{0}$

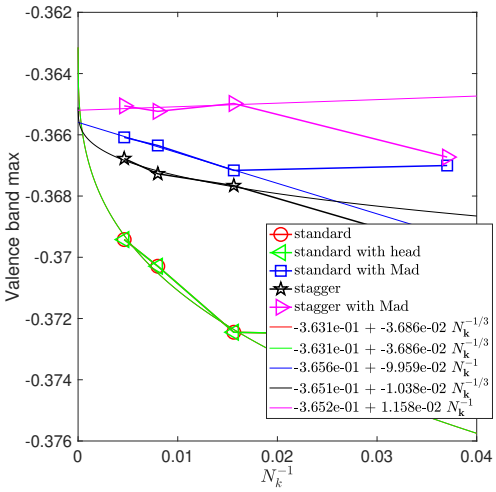


(b) CBM at $\mathbf{k} = \mathbf{0}$

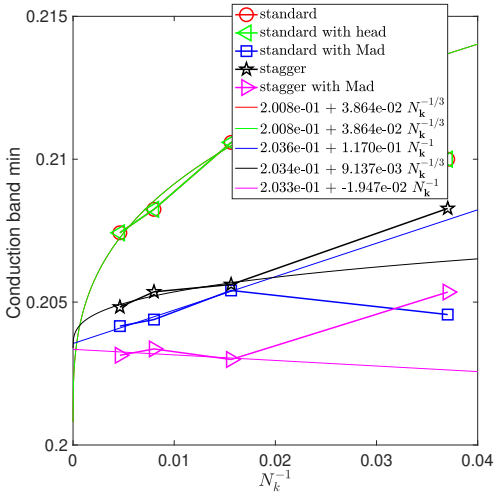


(c) Band gap at $\mathbf{k} = \mathbf{0}$

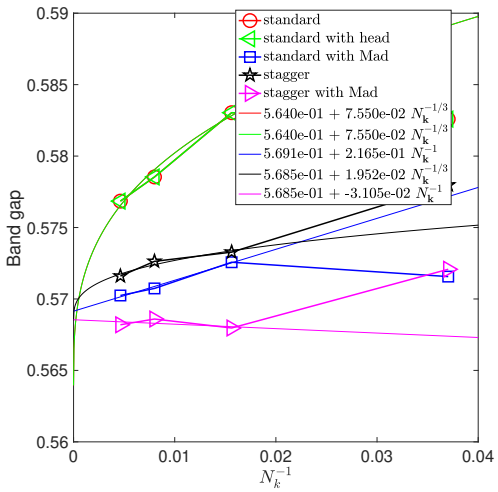
Setting 2 only include Σ^c



(a) VBM at $\mathbf{k} = \mathbf{0}$

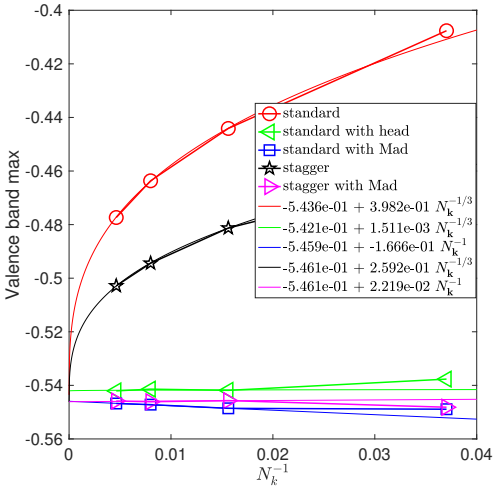


(b) CBM at $\mathbf{k} = \mathbf{0}$

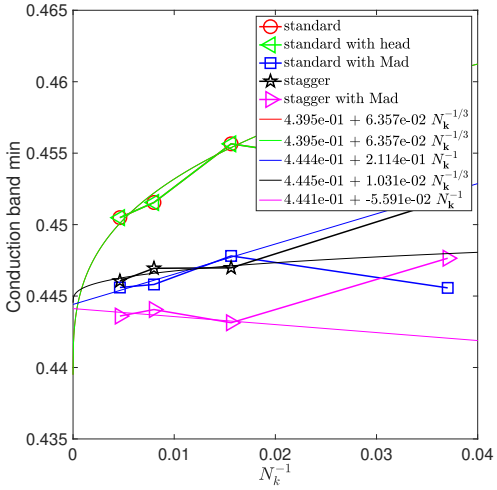


(c) Band gap at $\mathbf{k} = \mathbf{0}$

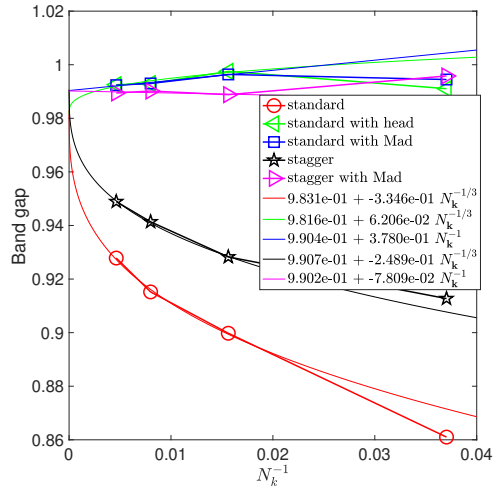
Setting 3 complete linearized G_0W_0 with $\Sigma^c + \Sigma^x - V_{xc}$



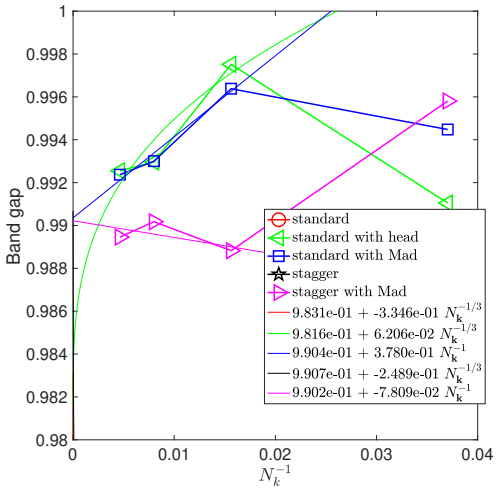
(a) VBM at $\mathbf{k} = \mathbf{0}$



(b) CBM at $\mathbf{k} = \mathbf{0}$



(c) Band gap at $\mathbf{k} = \mathbf{0}$



(d) ZOOMIN Band gap at $\mathbf{k} = \mathbf{0}$